

ALFAAA15484

## Tetra-n-butylammonium iodide

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

产品说明:  
 Product Description: 四正丁基碘化铵  
 Tetra-n-butylammonium iodide

Cat No. : A15484  
 CAS No 311-28-4  
 Molecular Formula C16 H36 I N

Supplier Avocado Research Chemicals Ltd.  
 (Part of Thermo Fisher Scientific)  
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Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

### SECTION 2. HAZARD IDENTIFICATION

Physical State  
 Powder Solid

Appearance  
 Off-white

Odor  
 Odorless

#### Emergency Overview

Harmful if swallowed. Causes skin irritation. Causes serious eye irritation. May cause respiratory irritation. Sensitivity to light.  
 Hygroscopic.

#### Classification of the substance or mixture

|                                                    |            |
|----------------------------------------------------|------------|
| Acute Oral Toxicity                                | Category 4 |
| Skin Corrosion/Irritation                          | Category 2 |
| Serious Eye Damage/Eye Irritation                  | Category 2 |
| Specific target organ toxicity - (single exposure) | Category 3 |

#### Label Elements



Signal Word

Warning

## Tetra-n-butylammonium iodide

**Hazard Statements**

H302 - Harmful if swallowed  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H335 - May cause respiratory irritation

**Precautionary Statements****Prevention**

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P330 - Rinse mouth  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Hygroscopic.

**Health Hazards**

Harmful if swallowed. Causes skin irritation. Causes serious eye irritation. May cause respiratory irritation.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Other Hazards**

Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                 | CAS No   | Weight % |
|---------------------------|----------|----------|
| Tetrabutylammonium iodide | 311-28-4 | >95      |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention if symptoms occur.

**Inhalation**

Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Get medical attention.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms and effects**

## Tetra-n-butylammonium iodide

No information available.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation. Avoid contact with skin, eyes or clothing.

**Environmental Precautions**

Avoid release to the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

**Storage**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

**SAFETY DATA SHEET****Tetra-n-butylammonium iodide**

| Component                 | ACGIH TLV                           | OSHA PEL | NIOSH | The United Kingdom | European Union |
|---------------------------|-------------------------------------|----------|-------|--------------------|----------------|
| Tetrabutylammonium iodide | TWA: 0.01 mg/m <sup>3</sup><br>Skin |          |       | -                  |                |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

**Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Small scale/Laboratory use** Maintain adequate ventilation

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**

|                       |                          |
|-----------------------|--------------------------|
| <b>Appearance</b>     | Off-white                |
| <b>Physical State</b> | Powder Solid             |
| <b>Odor</b>           | Odorless                 |
| <b>Odor Threshold</b> | No data available        |
| <b>pH</b>             | No information available |

**SAFETY DATA SHEET****Tetra-n-butylammonium iodide**

|                                                |                               |                                          |
|------------------------------------------------|-------------------------------|------------------------------------------|
| <b>Melting Point/Range</b>                     | 141 - 150 °C / 285.8 - 302 °F |                                          |
| <b>Softening Point</b>                         | No data available             |                                          |
| <b>Boiling Point/Range</b>                     | No information available      |                                          |
| <b>Flash Point</b>                             | No information available      | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable                | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available      |                                          |
| <b>Explosion Limits</b>                        | No data available             |                                          |
| <b>Vapor Pressure</b>                          | No data available             |                                          |
| <b>Vapor Density</b>                           | Not applicable                | Solid                                    |
| <b>Specific Gravity / Density</b>              | No data available             |                                          |
| <b>Bulk Density</b>                            | No data available             |                                          |
| <b>Water Solubility</b>                        | <5%                           |                                          |
| <b>Solubility in other solvents</b>            | No information available      |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                               |                                          |
| <b>Component</b>                               | <b>log Pow</b>                |                                          |
| Tetrabutylammonium iodide                      | 0.869                         |                                          |
| <b>Autoignition Temperature</b>                | No data available             |                                          |
| <b>Decomposition Temperature</b>               | No data available             |                                          |
| <b>Viscosity</b>                               | Not applicable                | Solid                                    |
| <b>Explosive Properties</b>                    | No information available      |                                          |
| <b>Oxidizing Properties</b>                    | No information available      |                                          |
| <b>Molecular Formula</b>                       | C16 H36 I N                   |                                          |
| <b>Molecular Weight</b>                        | 369.36                        |                                          |

**SECTION 10. STABILITY AND REACTIVITY**

|                                 |                                                                                                              |
|---------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Stability</b>                | Hygroscopic, Light sensitive.                                                                                |
| <b>Hazardous Reactions</b>      | No information available.                                                                                    |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur.                                                                     |
| <b>Conditions to Avoid</b>      | Avoid dust formation. Incompatible products. Excess heat. Exposure to light. Exposure to moist air or water. |
| <b>Materials to avoid</b>       | Strong oxidizing agents.                                                                                     |

**Hazardous Decomposition Products** Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen iodide.

**SECTION 11. TOXICOLOGICAL INFORMATION****Product Information****(a) acute toxicity;**

| Component                 | LD50 Oral                 | LD50 Dermal | LC50 Inhalation |
|---------------------------|---------------------------|-------------|-----------------|
| Tetrabutylammonium iodide | LD50 = 1990 mg/kg ( Rat ) |             |                 |

**(b) skin corrosion/irritation;** Category 2

**(c) serious eye damage/irritation;** Category 2

**(d) respiratory or skin sensitization;**

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | No data available |
| <b>Skin</b>        | No data available |

## Tetra-n-butylammonium iodide

|                                            |                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| (e) germ cell mutagenicity;                | No data available                                                                                                 |
| (f) carcinogenicity;                       | No data available<br>There are no known carcinogenic chemicals in this product                                    |
| (g) reproductive toxicity;                 | No data available                                                                                                 |
| (h) STOT-single exposure;                  | Category 3                                                                                                        |
| Results / Target organs                    | Respiratory system                                                                                                |
| (i) STOT-repeated exposure;                | No data available                                                                                                 |
| Target Organs                              | No information available.                                                                                         |
| (j) aspiration hazard;                     | Not applicable<br>Solid                                                                                           |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated. See actual entry in RTECS for complete information |
| Symptoms / effects, both acute and delayed | No information available                                                                                          |

## SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity effects Do not empty into drains.

Persistence and Degradability No information available

Bioaccumulative Potential No information available

| Component                 | log Pow | Bioconcentration factor (BCF) |
|---------------------------|---------|-------------------------------|
| Tetrabutylammonium iodide | 0.869   | No data available             |

Mobility in soil No information available

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors  
Persistent Organic Pollutant This product does not contain any known or suspected substance  
Ozone Depletion Potential This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point.

Other Information Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

## SECTION 14. TRANSPORT INFORMATION

## Tetra-n-butylammonium iodide

**Road and Rail Transport** Not Regulated**IMDG/IMO** Not regulated**IATA** Not regulated**Special Precautions for User** No special precautions required**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component                 | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|---------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Tetrabutylammonium iodide | -                                                   | -                                       | X    | X     | 206-220-5 | X    | X   | X     | -    |      | X    | KE-33274 |

**National Regulations****SECTION 16. OTHER INFORMATION**

**Prepared By** Health, Safety and Environmental Department  
**Creation Date** 08-Oct-2014  
**Revision Date** 22-Apr-2024  
**Revision Summary** New emergency telephone response service provider.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Legend****CAS** - Chemical Abstracts Service**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**NZIoC** - New Zealand Inventory of Chemicals**WEL** - Workplace Exposure Limit**ACGIH** - American Conference of Governmental Industrial Hygienists**TWA** - Time Weighted Average**IARC** - International Agency for Research on Cancer

## Tetra-n-butylammonium iodide

**DNEL** - Derived No Effect Level  
**RPE** - Respiratory Protective Equipment  
**LC50** - Lethal Concentration 50%  
**NOEC** - No Observed Effect Concentration  
**PBT** - Persistent, Bioaccumulative, Toxic

**PNEC** - Predicted No Effect Concentration  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road  
**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**ATE** - Acute Toxicity Estimate  
**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**